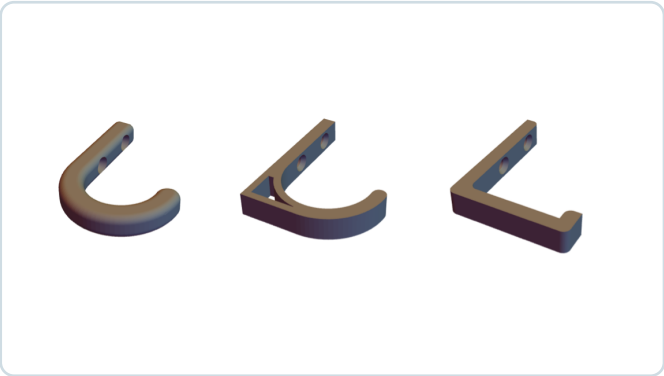
ILLUSTRATED RESEARCH DATASET

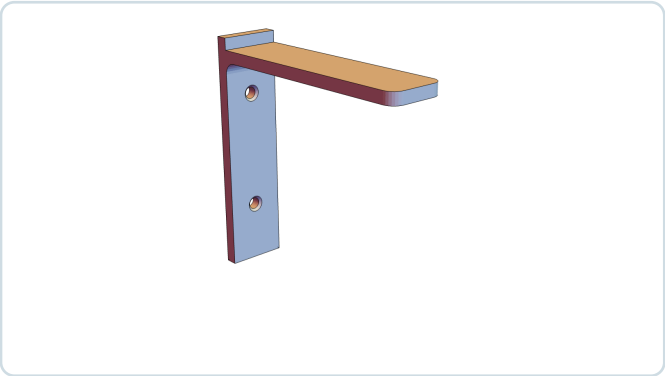
100 tasks in CAD + FEM

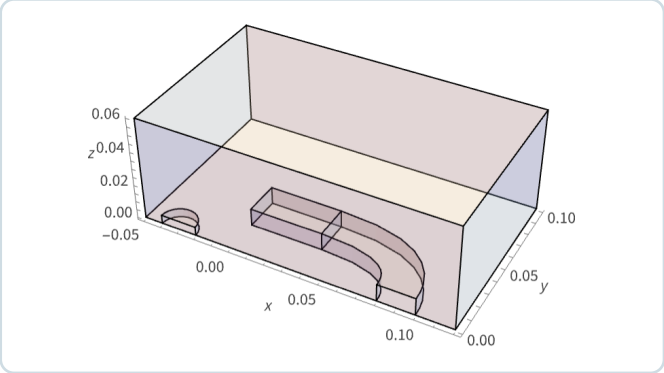
Engineering geometry. Source-defined physics.

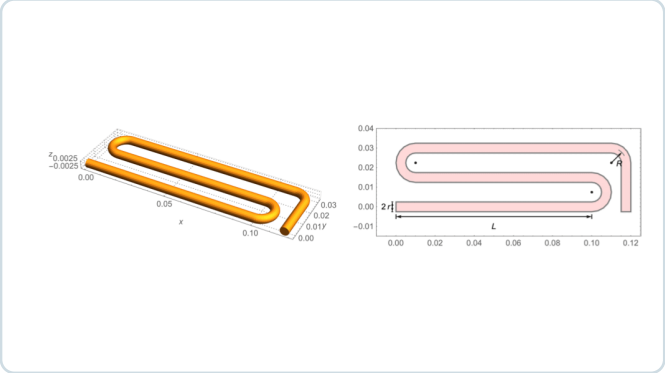
100 cases / 25 geometry families / 9 covered categories

13 hard | 55 complex | 32 hardest









WHAT THIS CATALOG CONTAINS

All 100 task specifications, source geometry images, case selectors, expected outputs and clickable references. Parameter variants share parts: this is not a collection of 100 distinct CAD geometries.

STATUS AND SCOPE

These are open-book reproduction and generation tasks. Difficulty labels are proposed. No gold answers or model failures have been validated. Accepted cases currently come from Wolfram; the MFEM crawl yielded no selected qualifying case. System Physics remains uncovered.

The drawings are source geometry views, engineering cross-sections or part meshes. Solution-field images and graph plots are excluded. Source images belong to their respective owners.

Shared task requirements

1. Reconstruct the CAD

Deliver a reproducible CAD construction and STEP geometry, or planar CAD where the source is 2D. Include the boundary/material region map. Preserve source dimensions, units, fillets and model dimensionality.

2. Reproduce the FEM problem

Deliver a mesh, FEM implementation and results.json. Preserve the constitutive law, loads, boundary and initial conditions. The individual task page specifies the analysis and required results.

3. Resolve the source inputs

Use the full downloadable code-cell URLs in sources.json, not abbreviated screen-reader captions. Cell indices in this catalog are zero-based HTML input order, not Mathematica In[] labels. Resolve prerequisite definitions from the same page.

4. Record dependencies

Resolve and record material-entity values, runtime defaults and installation support files. Report missing dependencies instead of guessing. Examples include the muffler STL files, bracket WXF geometry and condenser meteorological data. These assets are not bundled as solved task inputs.

5. Verify the prediction

Include mesh convergence and appropriate balance/residual checks. CAD export and verification requirements are evaluation instructions authored for this dataset, not measurements claimed from the source.

6. Avoid evaluation leakage

Split by split_group, not by individual row. The 21 conservative split groups keep related hooks, mufflers, perforated plates and all bracket variants together. Source tutorials contain worked solutions; an image-only evaluation requires a separate input/reference split.

Source and validation record

28 unique source images and 1,003 full source-code cells were traced and checked. Structural/provenance checks passed; no FEM solve was performed. The JSONL remains the machine-readable source of record. Legacy numerical assumptions were replaced by official-source bindings.

Task index 1-25

Select any task title to jump to its illustrated specification.

001 Disc brake: axisymmetric friction heating	complex
002 Shrink-fit rod and frame: transient assembly clearance	complex
003 Etched anemometer: stationary thermal sensing	hardest
004 Branched spherical chamber: ramped transient flow	complex
005 Printed hook A: filleted curved design under load	hardest
006 Printed hook B: pocketed reinforced design under load	complex
007 Printed hook C: filleted angular design under load	complex
008 Dogbone: fitted Yeoh model with stiff clamp regions	hardest
009 Perforated biaxial specimen: finite-strain loading	hardest
010 Neck-and-cavity resonator: forced frequency response	complex
011 Coil pair: axisymmetric magnetostatic field	complex
012 MEMS actuator: electrostatic and structural coupling	hardest
013 Gasket: coupled diffusion, heat and hyperelastic swelling	hardest
014 Passive condenser: wind, radiation and dew collection	hardest
015 Bookshelf bracket: mounted eigenmodes	complex
016 Bookshelf bracket: unconstrained eigenmodes	complex
017 L-bracket: fillet stress convergence	hard
018 Steel block with aluminum insert: thermal expansion	complex
019 Plate with bore: 3D tensile verification	hard
020 Perforated plate: plastic loading and unloading history	hardest
021 Horseshoe permanent magnet and coin	hardest
022 Looped manifold: non-Newtonian flow	complex
023 Internally heated tube: axisymmetric verification	hard
024 Tungsten heating coil: current and Joule heat	complex
025 Disc brake: transient thermal stress	hardest

Task index 26-50

Select any task title to jump to its illustrated specification.

026 Internally heated tube: full 3D verification	complex
027 Dogbone: fitted Neo-Hookean 3D tensile experiment	hardest
028 Bookshelf bracket: source force sweep, 100 N	hard
029 Bookshelf bracket: source force sweep, 200 N	hard
030 Bookshelf bracket: source force sweep, 300 N	hard
031 Bookshelf bracket: source force sweep, 400 N	hard
032 Bookshelf bracket: source force sweep, 500 N	hard
033 Bookshelf bracket: source force sweep, 600 N	hard
034 Bookshelf bracket: source force sweep, 700 N	hard
035 Bookshelf bracket: source force sweep, 800 N	hard
036 Bookshelf bracket: source force sweep, 900 N	hard
037 Bookshelf bracket: source force sweep, 1000 N	hard
038 Bookshelf bracket: source harmonic case 01	complex
039 Bookshelf bracket: source harmonic case 02	complex
040 Bookshelf bracket: source harmonic case 03	complex
041 Bookshelf bracket: source harmonic case 04	complex
042 Bookshelf bracket: source harmonic case 05	complex
043 Bookshelf bracket: source harmonic case 06	complex
044 Bookshelf bracket: source harmonic case 07	complex
045 Bookshelf bracket: source harmonic case 08	complex
046 Bookshelf bracket: source harmonic case 09	complex
047 Bookshelf bracket: source harmonic case 10	complex
048 Bookshelf bracket: source harmonic case 11	complex
049 Bookshelf bracket: source harmonic case 12	complex
050 Bookshelf bracket: source harmonic case 13	complex

Task index 51-75

Select any task title to jump to its illustrated specification.

051 Bookshelf bracket: source harmonic case 14	complex
052 Bookshelf bracket: source harmonic case 15	complex
053 Bookshelf bracket: source harmonic case 16	complex
054 Bookshelf bracket: source harmonic case 17	complex
055 Bookshelf bracket: source harmonic case 18	complex
056 Bookshelf bracket: source harmonic case 19	complex
057 Bookshelf bracket: source harmonic case 20	complex
058 Bookshelf bracket: source harmonic case 21	complex
059 Bookshelf bracket: source harmonic case 22	complex
060 Bookshelf bracket: source harmonic case 23	complex
061 Bookshelf bracket: source harmonic case 24	complex
062 Bookshelf bracket: source harmonic case 25	complex
063 Bookshelf bracket: source harmonic case 26	complex
064 Bookshelf bracket: source harmonic case 27	complex
065 Bookshelf bracket: source harmonic case 28	complex
066 Bookshelf bracket: source harmonic case 29	complex
067 Bookshelf bracket: source harmonic case 30	complex
068 Bookshelf bracket: source harmonic case 31	complex
069 Bookshelf bracket: transient sinusoidal loading	hardest
070 Bookshelf bracket: source Poisson-ratio comparison	complex
071 Classical muffler: 100 Hz	complex
072 Classical muffler: 200 Hz	complex
073 Classical muffler: 300 Hz	complex
074 Classical muffler: 400 Hz	complex
075 Classical muffler: 500 Hz	complex

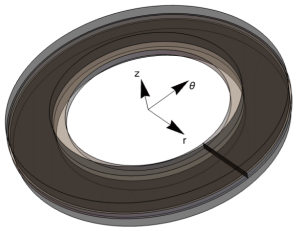
Task index 76-100

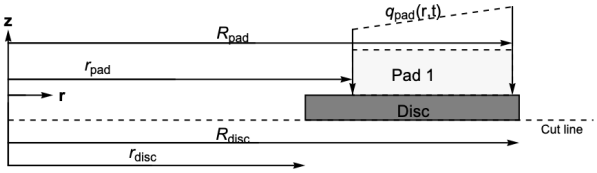
Select any task title to jump to its illustrated specification.

076 Classical muffler: 600 Hz	complex
077 Classical muffler: 700 Hz	complex
078 Classical muffler: 800 Hz	complex
079 Classical muffler: 900 Hz	complex
080 Classical muffler: 1000 Hz	complex
081 Perforated muffler: 100 Hz	hardest
082 Perforated muffler: 200 Hz	hardest
083 Perforated muffler: 300 Hz	hardest
084 Perforated muffler: 400 Hz	hardest
085 Perforated muffler: 500 Hz	hardest
086 Perforated muffler: 600 Hz	hardest
087 Perforated muffler: 700 Hz	hardest
088 Perforated muffler: 800 Hz	hardest
089 Perforated muffler: 900 Hz	hardest
090 Perforated muffler: 1000 Hz	hardest
091 Padded muffler: 100 Hz	hardest
092 Padded muffler: 200 Hz	hardest
093 Padded muffler: 300 Hz	hardest
094 Padded muffler: 400 Hz	hardest
095 Padded muffler: 500 Hz	hardest
096 Padded muffler: 600 Hz	hardest
097 Padded muffler: 700 Hz	hardest
098 Padded muffler: 800 Hz	hardest
099 Padded muffler: 900 Hz	hardest
100 Padded muffler: 1000 Hz	hardest

Disc brake: axisymmetric friction heating

COMPLEX (proposed) | Family: brake-disc | Split: brake-disc





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the disc and pad-track partition. Reproduce the source axisymmetric transient thermal analysis, including its actual flux and natural boundary conditions.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Temperature field and hottest-point temperature at the source end time.

SOURCE BINDING

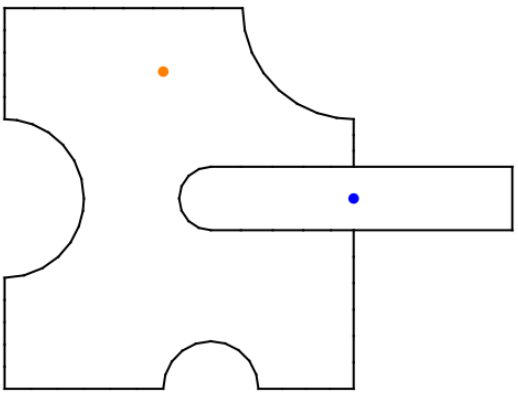
[01] Thermal and Structural Analysis of a Disc Brake

Input cells: 0-17.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Shrink-fit rod and frame: transient assembly clearance

COMPLEX (proposed) | Family: shrink-fit-assembly | Split: shrink-fit-assembly



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct both contacting regions. Reproduce the source transient heat problem and its clearance calculation.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Temperatures of both components and assembly-clearance history.

SOURCE BINDING

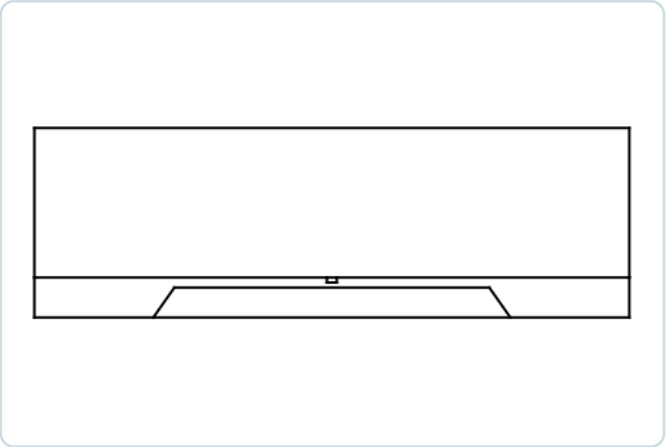
[02] Shrink Fitting of Assembly

Input cells: 0-28.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Etched anemometer: stationary thermal sensing

HARDEST (proposed) | Family: anemometer-chip | Split: anemometer-chip



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the etched membrane, heater, cavity and channel with original region and boundary markers. Reproduce the stationary source model; resolve the source length scale explicitly.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Temperature field, heater temperature and region/boundary marker map.

SOURCE BINDING

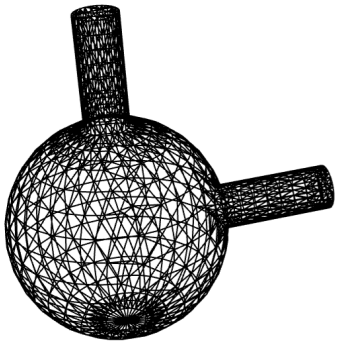
[03] Contactless Anemometer

Input cells: 0-33.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Branched spherical chamber: ramped transient flow

COMPLEX (proposed) | Family: branched-chamber | Split: branched-chamber



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the connected chamber and pipes. Use the source transient Navier-Stokes equations, ramped inlet and initial conditions; preserve the mixed interpolation orders.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Velocity and pressure fields at the final source time, inlet/outlet fluxes and continuity residual.

SOURCE BINDING

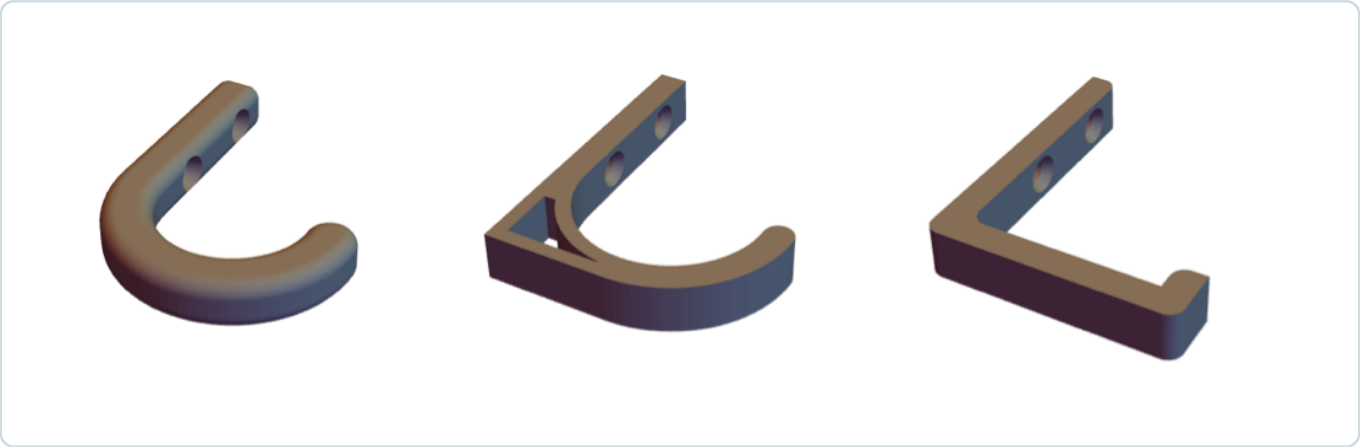
[04] Laminar Flow

Input cells: 0-1, 55-69.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Printed hook A: filleted curved design under load

HARDEST (proposed) | Family: hook-a | Split: printed-hooks



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct hook1 with all source fillets and countersunk fixation holes. Apply the material calibration, infill choice, supports and load of the source mechanical comparison.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Displacement and stress fields, resultant support force, and comparison quantities for hook1.

SOURCE BINDING

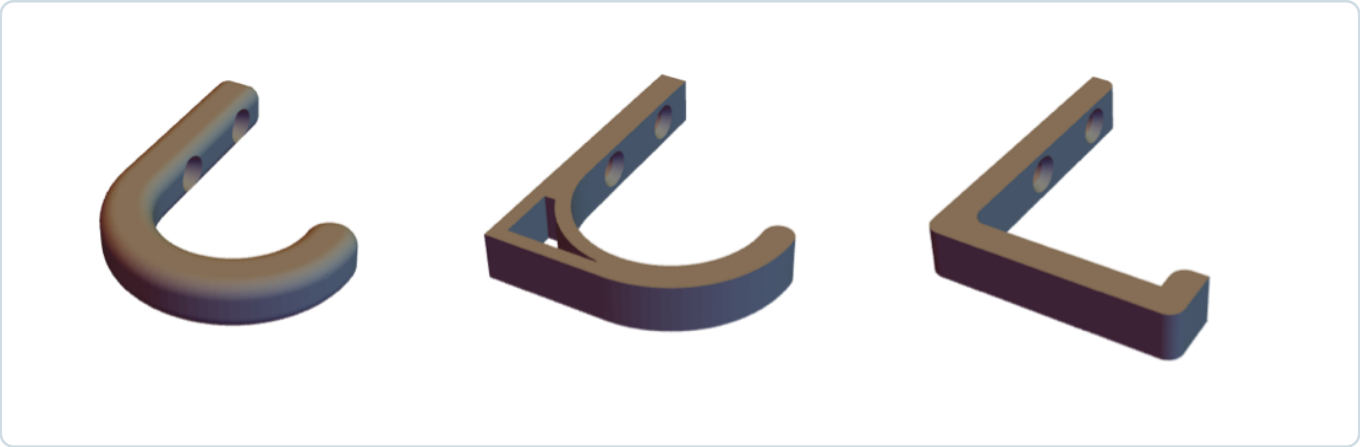
[05] 3D Print and Mechanical Design

Input cells: 0-67.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Printed hook B: pocketed reinforced design under load

COMPLEX (proposed) | Family: hook-b | Split: printed-hooks



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct hook2 including its pocket and fixation holes. Apply the source mechanical comparison setup and its calibrated material.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Displacement and stress fields and comparison quantities for hook2.

SOURCE BINDING

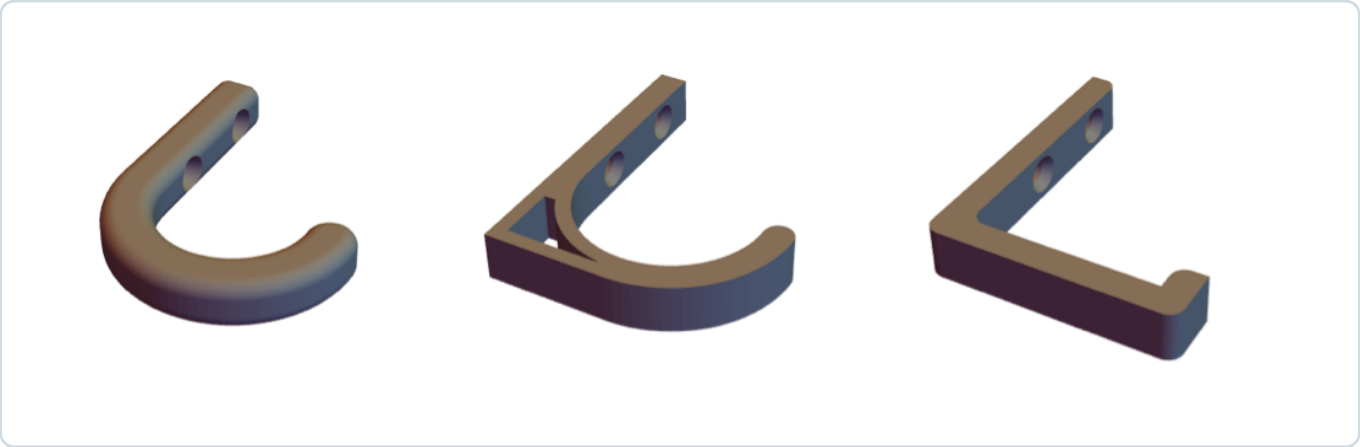
[05] 3D Print and Mechanical Design

Input cells: 0-67.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Printed hook C: filleted angular design under load

COMPLEX (proposed) | Family: hook-c | Split: printed-hooks



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct hook3 including the five specified edge fillets and fixation holes. Preserve the source load and calibrated material.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Displacement and stress fields and comparison quantities for hook3.

SOURCE BINDING

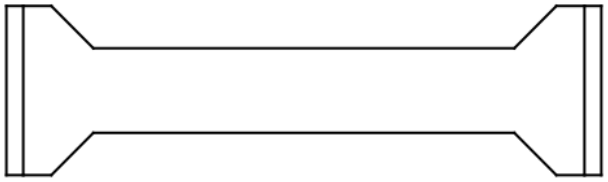
[05] 3D Print and Mechanical Design

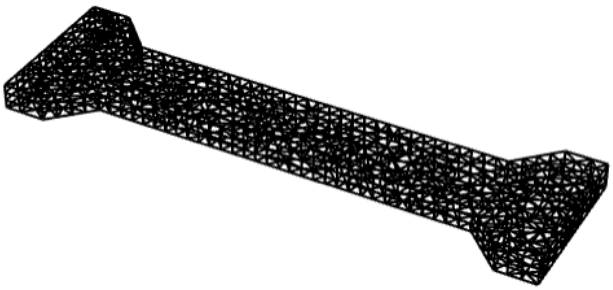
Input cells: 0-67.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Dogbone: fitted Yeoh model with stiff clamp regions

HARDEST (proposed) | Family: dogbone | Split: dogbone





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the specimen and clamps. Reproduce the source fitted Yeoh FEM experiment, including the actual 2D model, thickness, clamp material and continuation schedule.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Full force-extension history and source strain/stress comparison quantities.

SOURCE BINDING

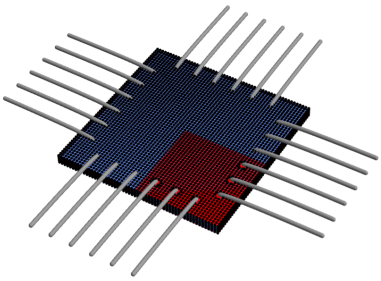
[06] Hyperelastic Model Comparison

Input cells: 0-74.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Perforated biaxial specimen: finite-strain loading

HARDEST (proposed) | Family: biaxial-specimen | Split: biaxial-specimen



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the quarter specimen and suture-hole boundaries. Reproduce the source displacement-controlled hyperelastic experiment and symmetry reduction.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Reaction forces, stress and displacement through the source loading schedule.

SOURCE BINDING

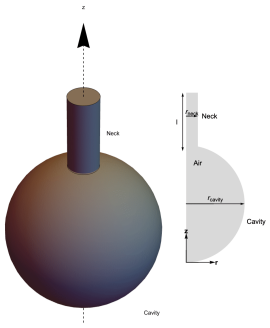
[07] Biaxial Tensile Test of Hyperelastic Tissue

Input cells: 0-27.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Neck-and-cavity resonator: forced frequency response

COMPLEX (proposed) | Family: helmholtz-resonator | Split: helmholtz-resonator



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the resonator air volume. Reproduce the source pressure-driven axisymmetric frequency sweep and frequency-dependent mesh sizing.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Complex pressure responses and cavity response over the source sweep; identify resonant peaks.

SOURCE BINDING

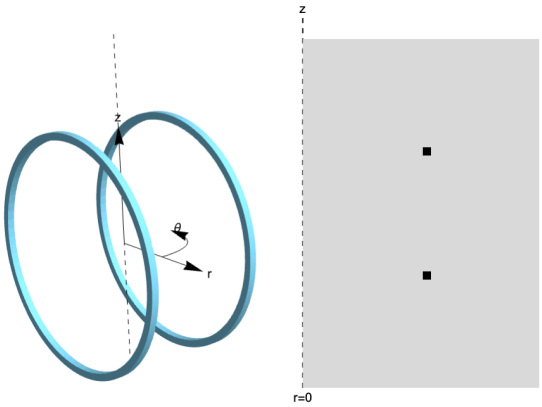
[08] Helmholtz Resonator

Input cells: 0-30.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Coil pair: axisymmetric magnetostatic field

COMPLEX (proposed) | Family: coil-pair | Split: coil-pair



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct both windings and the exterior air region. Reproduce the source magnetostatic setup and its axis treatment.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Magnetic vector potential, flux density and center-field value with its limiting treatment on the axis.

SOURCE BINDING

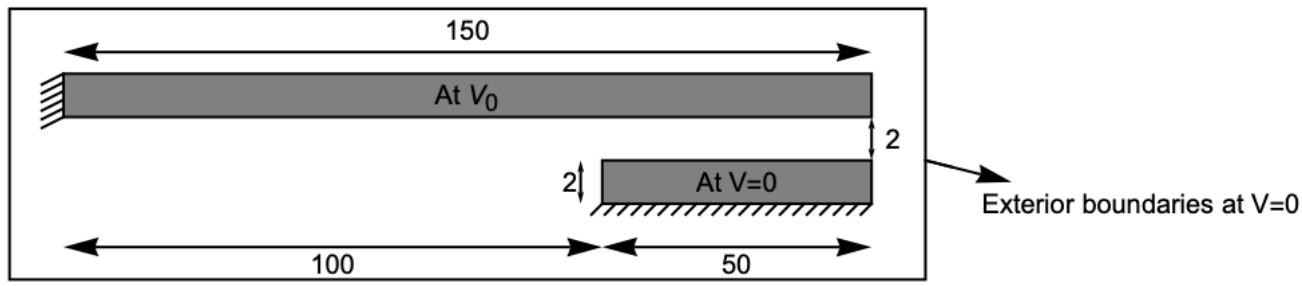
[09] Quasistatic Magnetic Fields

Input cells: 0-54.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

MEMS actuator: electrostatic and structural coupling

HARDEST (proposed) | Family: mems-beam | Split: mems-beam



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the beam, electrode and air regions. Follow the source electrostatic-to-structural coupling and displacement iteration.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Potential, beam displacement and converged electrostatic load, with iteration history.

SOURCE BINDING

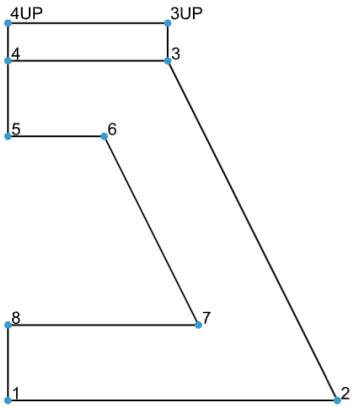
[10] Electrostatically Actuated MEMS Device

Input cells: 0-47.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Gasket: coupled diffusion, heat and hyperelastic swelling

HARDEST (proposed) | Family: hygroscopic-gasket | Split: hygroscopic-gasket



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the gasket and loading plate. Keep the source diffusion, thermal and mechanical coupling and continuation; retain deformation-dependent transport coefficients.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Concentration, temperature and displacement fields through the complete continuation.

SOURCE BINDING

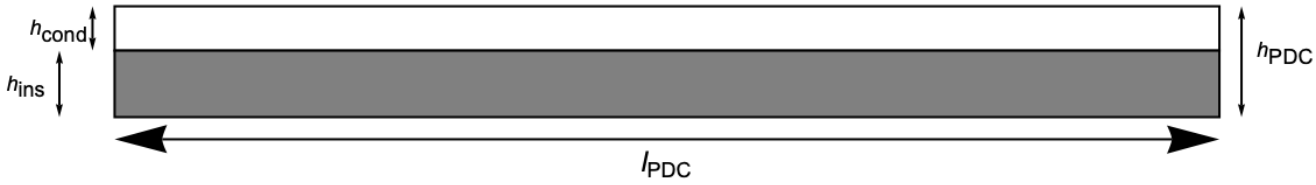
[11] Hygroscopic Swelling

Input cells: 0-74.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Passive condenser: wind, radiation and dew collection

HARDEST (proposed) | Family: dew-condenser | Split: dew-condenser



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the condenser, insulation, supports and flow domain. Preserve the source meteorological inputs, sky radiation, heat transfer and condensation calculation.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Flow and temperature fields, surface-temperature history and predicted condensed-water quantity.

SOURCE BINDING

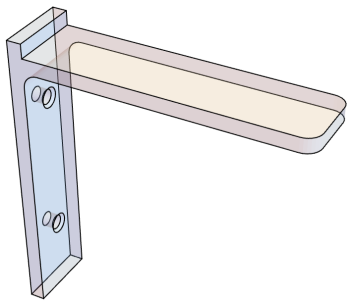
[12] Water Condensation of Passive Dew Condensers

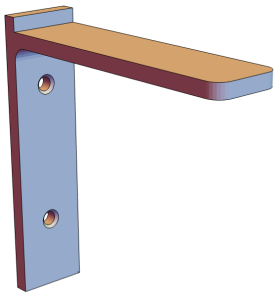
Input cells: 0-168.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: mounted eigenmodes

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the complete bracket and mounting features. Use the source material entity and mounting conditions in its six-mode eigenanalysis.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Six source-convention eigenfrequencies and normalized mode fields.

SOURCE BINDING

[13] Solid Mechanics

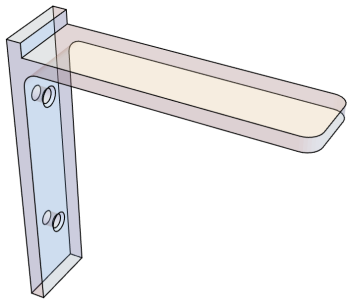
Input cells: 0-30, 82-98.

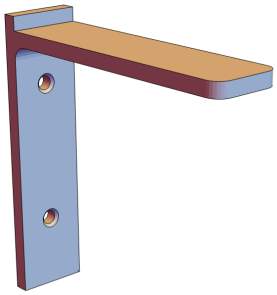
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: unconstrained eigenmodes

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the complete bracket. Reproduce the source unconstrained twelve-mode solve, retaining and identifying rigid-body modes.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Twelve eigenvalues, rigid-mode identification and elastic mode fields.

SOURCE BINDING

[13] Solid Mechanics

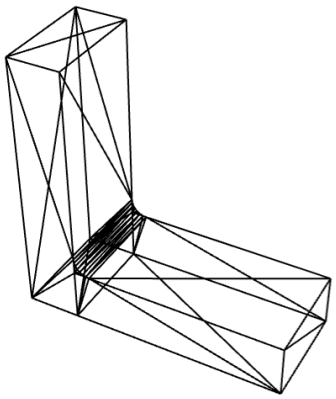
Input cells: 0-30, 82-102.

Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

L-bracket: fillet stress convergence

HARD (proposed) | Family: filleted-l-bracket | Split: filleted-l-bracket



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source filleted L-bracket. Reproduce its refined curved-boundary stress analysis and compare the documented mesh variants.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Mesh-dependent maximum von Mises stress and convergence evidence; distinguish the sharp-corner singularity from the filleted case.

SOURCE BINDING

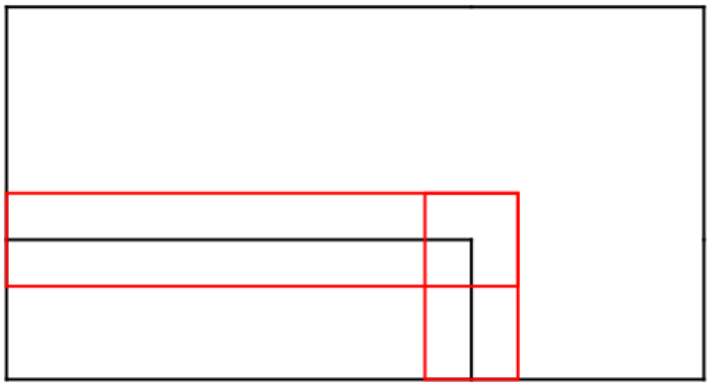
[13] Solid Mechanics

Input cells: 0-1, 547-563.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Steel block with aluminum insert: thermal expansion

COMPLEX (proposed) | Family: bimetal-insert | Split: bimetal-insert



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct both material regions and their shared interface. Reproduce the source plane-stress thermal-strain problem with original thickness and constraints.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Displacement, interface tractions and stress components.

SOURCE BINDING

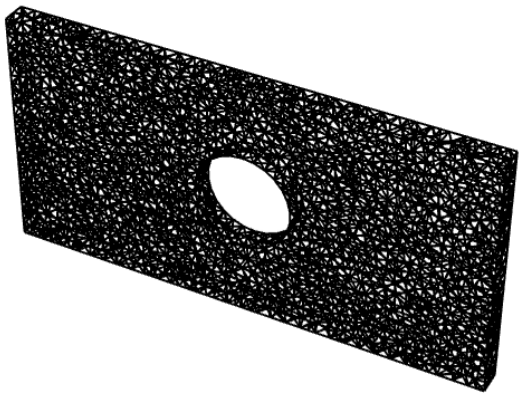
[13] Solid Mechanics

Input cells: 0-1, 286-301.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Plate with bore: 3D tensile verification

HARD (proposed) | Family: plate-hole-3d | Split: perforated-plates



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the perforated plate. Reproduce the source 3D verification case, including both documented equation formulations.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Source probe displacement/stress values and verification error against the source test target.

SOURCE BINDING

[15] Solid Mechanics Model Verification Tests

Input cells: 0-2, 161-175.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Perforated plate: plastic loading and unloading history

HARDEST (proposed) | Family: plate-hole-plastic | Split: perforated-plates

Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the quarter plate. Reproduce the source extended-plane-strain loading/unloading sequence without resetting plastic state between increments.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Displacement and plastic strain at every step, including residual deformation after unloading.

SOURCE BINDING

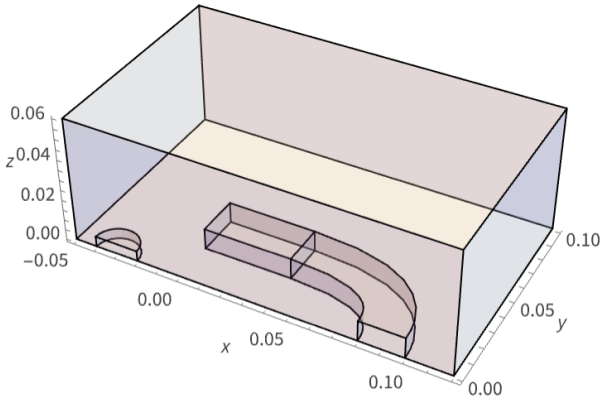
[16] Plasticity

Input cells: 0, 150-173.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Horseshoe permanent magnet and coin

HARDEST (proposed) | Family: magnet-coin | Split: magnet-coin



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the magnet segments, coin and exterior domain. Reproduce the source magnetization directions, material interfaces and scalar-potential FEM analysis.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Potential, flux-density field and source secondary magnetic quantities.

SOURCE BINDING

[17] Magnetostatics for Permanent Magnets

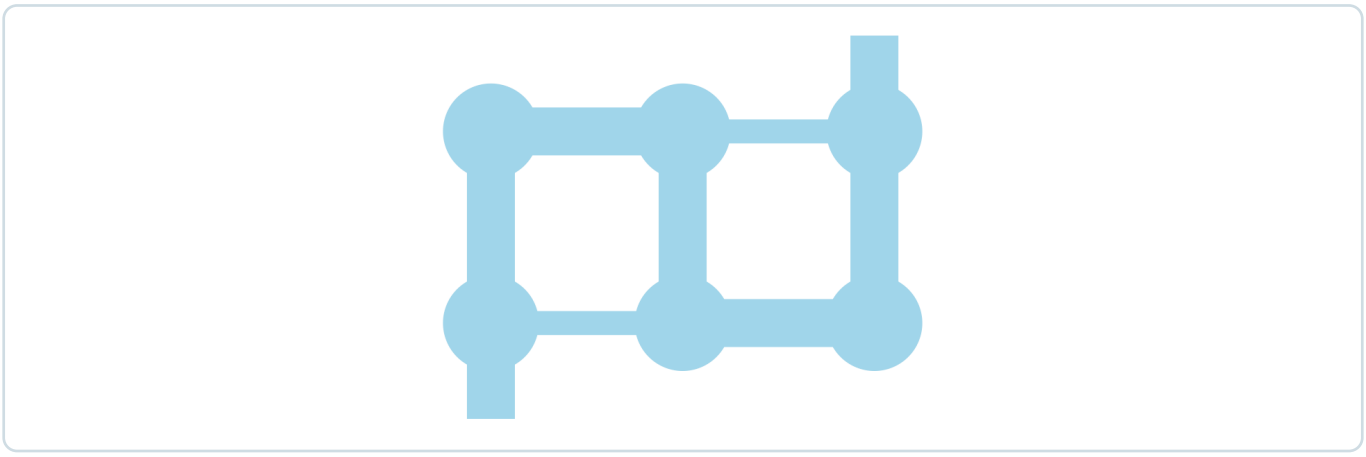
Input cells: 0-43.

Geometry construction: source [18]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Looped manifold: non-Newtonian flow

COMPLEX (proposed) | Family: looped-manifold | Split: looped-manifold



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the branched planar flow region. Reproduce the source power-law fluid, inlet and pressure reference, preserving the planar model and source units.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Velocity, pressure, apparent viscosity and viscous stress fields.

SOURCE BINDING

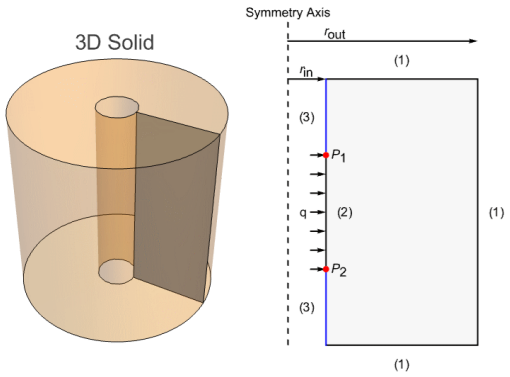
[04] Laminar Flow

Input cells: 0-1, 158-173.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Internally heated tube: axisymmetric verification

HARD (proposed) | Family: heated-tube | Split: heated-tube



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the tube cross-section and heated inner-wall band. Reproduce the source axisymmetric steady heat verification.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Temperature at the official probe and source verification residual.

SOURCE BINDING

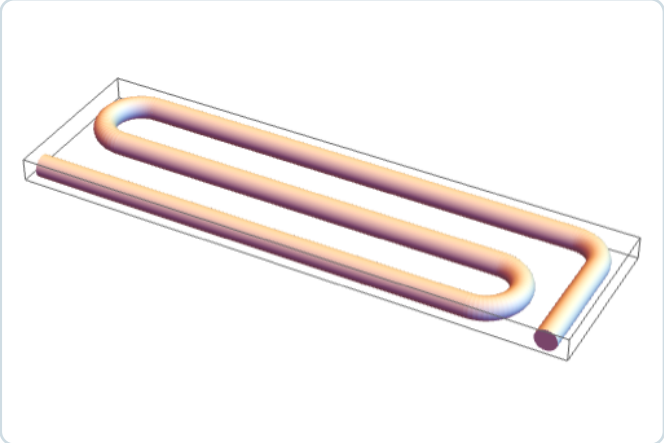
[19] Heat Transfer Model Verification Tests

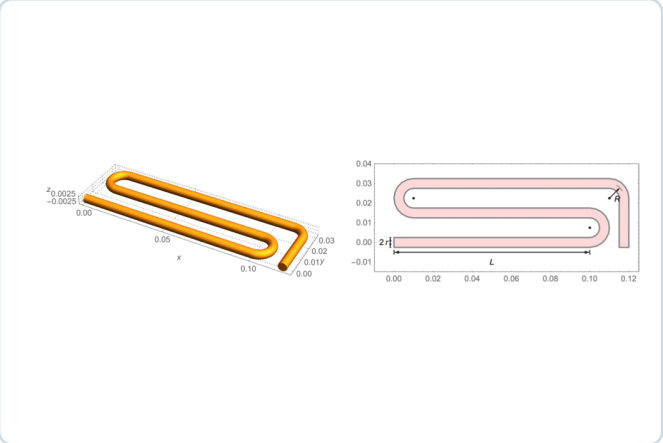
Input cells: 0, 36-46.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Tungsten heating coil: current and Joule heat

COMPLEX (proposed) | Family: heating-coil | Split: heating-coil





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the rounded wire coil and terminal faces. Reproduce the source electric-current and heat-transfer chain, including heat loss and source material properties.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Potential, current density, Joule source and final temperature field.

SOURCE BINDING

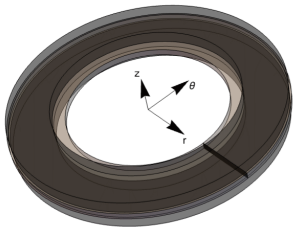
[20] Joule Heating of Tungsten Wire

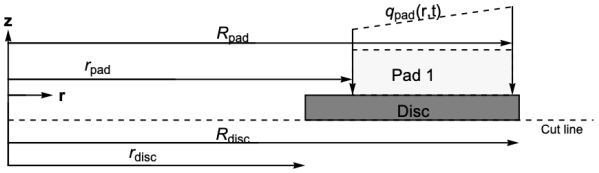
Input cells: 0-38.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Disc brake: transient thermal stress

HARDEST (proposed) | Family: brake-disc | Split: brake-disc





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source disc. Solve its thermal history followed by the documented axisymmetric structural analysis, carrying the spatial temperature field into each structural step.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Displacement and stress history with source temperature-to-stress transfer.

SOURCE BINDING

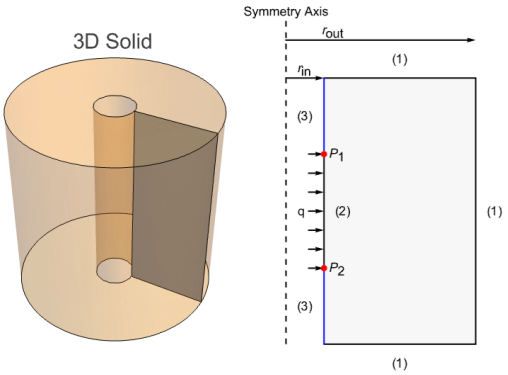
[01] Thermal and Structural Analysis of a Disc Brake

Input cells: 0-49.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Internally heated tube: full 3D verification

COMPLEX (proposed) | Family: heated-tube | Split: heated-tube



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Construct the full tube with the source inner-wall band. Reproduce the explicitly documented 3D thermal verification and compare its axisymmetric counterpart.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Temperature at the 3D source probe, source verification residual and agreement with the axisymmetric solution.

SOURCE BINDING

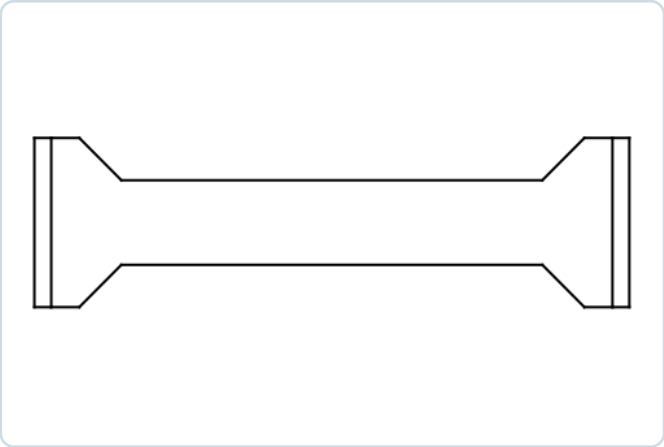
[19] Heat Transfer Model Verification Tests

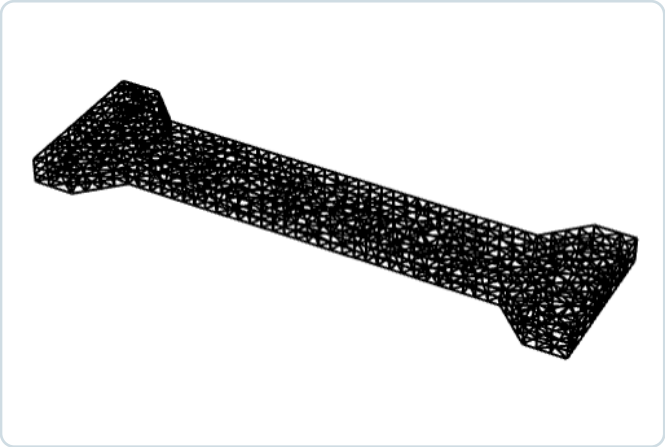
Input cells: 0, 36-51.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Dogbone: fitted Neo-Hookean 3D tensile experiment

HARDEST (proposed) | Family: dogbone | Split: dogbone





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the extruded dogbone and clamp extensions. Reproduce the source 3D Neo-Hookean experiment and its fitted material parameters and incremental force schedule.

CASE SELECTION

Documented analysis identified by the source cells below; preserve all prerequisites and any loading history.

REQUIRED RESULTS

Force-extension history, deformed geometry and strain field.

SOURCE BINDING

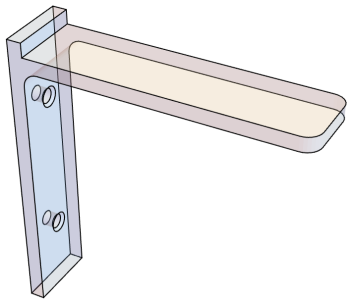
[06] Hyperelastic Model Comparison

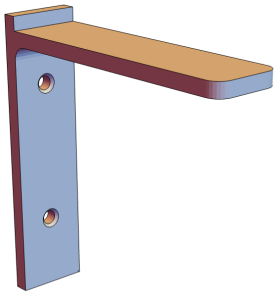
Input cells: 0-52.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source force sweep, 100 N

HARD (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket with fillets, chamfers and fixation holes. Solve this member of the published force sweep using the original material entity and supports.

CASE SELECTION

force = 100 N. Source sweep: Range[100, 1000, 100].

REQUIRED RESULTS

Displacement magnitude at the source sweep probe, displacement/stress fields and support reactions.

SOURCE BINDING

[13] Solid Mechanics

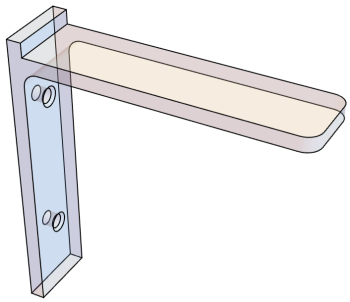
Input cells: 0-30, 103-108.

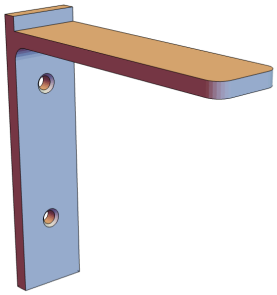
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source force sweep, 200 N

HARD (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket with fillets, chamfers and fixation holes. Solve this member of the published force sweep using the original material entity and supports.

CASE SELECTION

force = 200 N. Source sweep: Range[100, 1000, 100].

REQUIRED RESULTS

Displacement magnitude at the source sweep probe, displacement/stress fields and support reactions.

SOURCE BINDING

[13] Solid Mechanics

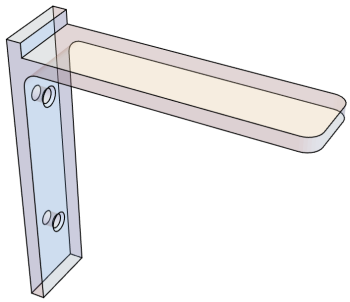
Input cells: 0-30, 103-108.

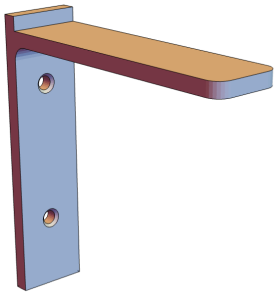
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source force sweep, 300 N

HARD (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket with fillets, chamfers and fixation holes. Solve this member of the published force sweep using the original material entity and supports.

CASE SELECTION

force = 300 N. Source sweep: Range[100, 1000, 100].

REQUIRED RESULTS

Displacement magnitude at the source sweep probe, displacement/stress fields and support reactions.

SOURCE BINDING

[13] Solid Mechanics

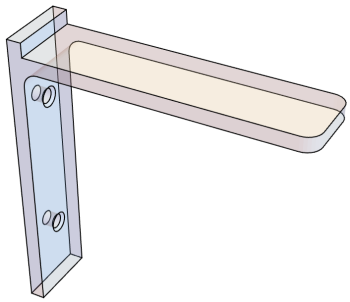
Input cells: 0-30, 103-108.

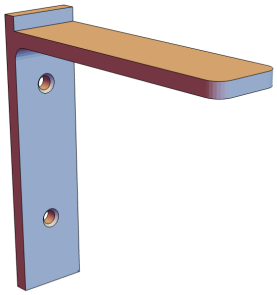
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source force sweep, 400 N

HARD (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket with fillets, chamfers and fixation holes. Solve this member of the published force sweep using the original material entity and supports.

CASE SELECTION

force = 400 N. Source sweep: Range[100, 1000, 100].

REQUIRED RESULTS

Displacement magnitude at the source sweep probe, displacement/stress fields and support reactions.

SOURCE BINDING

[13] Solid Mechanics

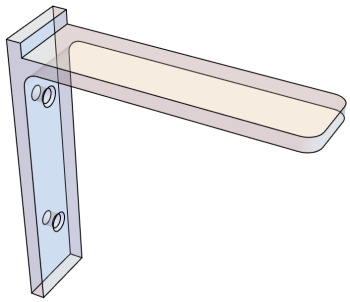
Input cells: 0-30, 103-108.

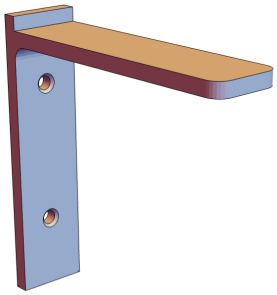
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source force sweep, 500 N

HARD (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket with fillets, chamfers and fixation holes. Solve this member of the published force sweep using the original material entity and supports.

CASE SELECTION

force = 500 N. Source sweep: Range[100, 1000, 100].

REQUIRED RESULTS

Displacement magnitude at the source sweep probe, displacement/stress fields and support reactions.

SOURCE BINDING

[13] Solid Mechanics

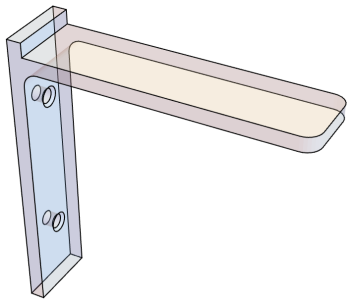
Input cells: 0-30, 103-108.

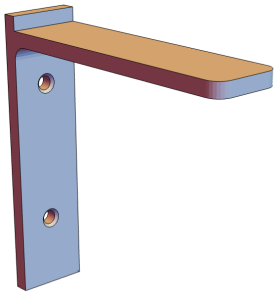
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source force sweep, 600 N

HARD (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket with fillets, chamfers and fixation holes. Solve this member of the published force sweep using the original material entity and supports.

CASE SELECTION

force = 600 N. Source sweep: Range[100, 1000, 100].

REQUIRED RESULTS

Displacement magnitude at the source sweep probe, displacement/stress fields and support reactions.

SOURCE BINDING

[13] Solid Mechanics

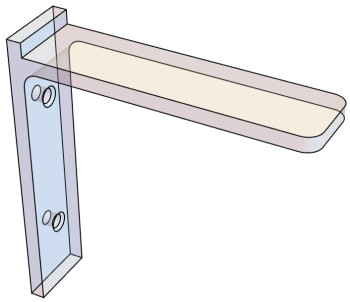
Input cells: 0-30, 103-108.

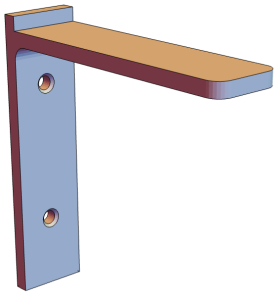
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source force sweep, 700 N

HARD (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket with fillets, chamfers and fixation holes. Solve this member of the published force sweep using the original material entity and supports.

CASE SELECTION

force = 700 N. Source sweep: Range[100, 1000, 100].

REQUIRED RESULTS

Displacement magnitude at the source sweep probe, displacement/stress fields and support reactions.

SOURCE BINDING

[13] Solid Mechanics

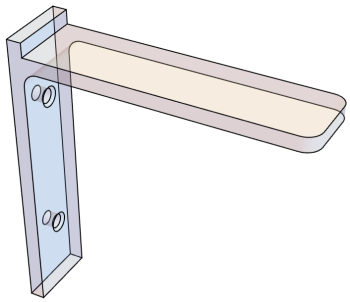
Input cells: 0-30, 103-108.

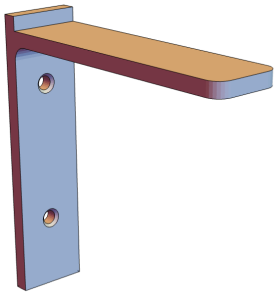
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source force sweep, 800 N

HARD (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket with fillets, chamfers and fixation holes. Solve this member of the published force sweep using the original material entity and supports.

CASE SELECTION

force = 800 N. Source sweep: Range[100, 1000, 100].

REQUIRED RESULTS

Displacement magnitude at the source sweep probe, displacement/stress fields and support reactions.

SOURCE BINDING

[13] Solid Mechanics

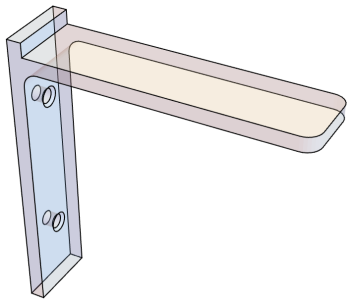
Input cells: 0-30, 103-108.

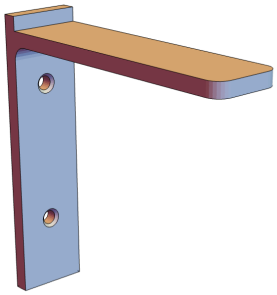
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source force sweep, 900 N

HARD (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket with fillets, chamfers and fixation holes. Solve this member of the published force sweep using the original material entity and supports.

CASE SELECTION

force = 900 N. Source sweep: Range[100, 1000, 100].

REQUIRED RESULTS

Displacement magnitude at the source sweep probe, displacement/stress fields and support reactions.

SOURCE BINDING

[13] Solid Mechanics

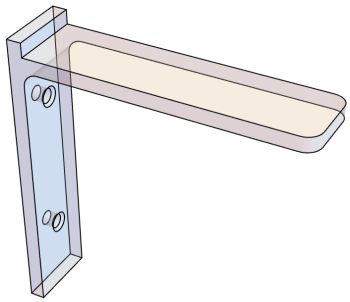
Input cells: 0-30, 103-108.

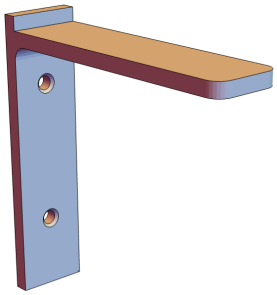
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source force sweep, 1000 N

HARD (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket with fillets, chamfers and fixation holes. Solve this member of the published force sweep using the original material entity and supports.

CASE SELECTION

force = 1000 N. Source sweep: Range[100, 1000, 100].

REQUIRED RESULTS

Displacement magnitude at the source sweep probe, displacement/stress fields and support reactions.

SOURCE BINDING

[13] Solid Mechanics

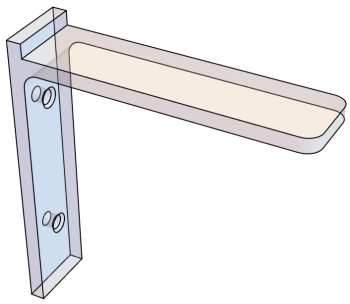
Input cells: 0-30, 103-108.

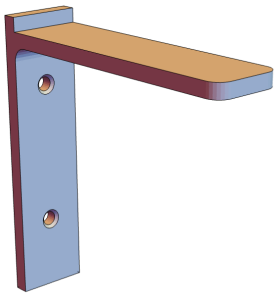
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 01

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(60/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

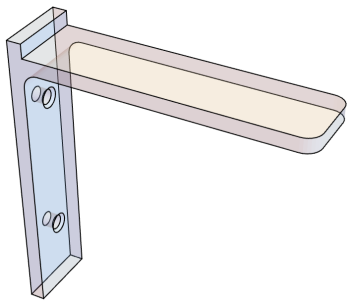
Input cells: 0-30, 114-123.

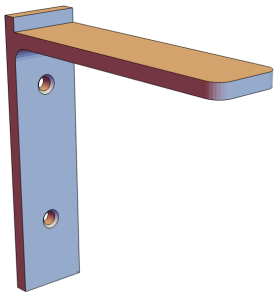
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 02

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(61/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

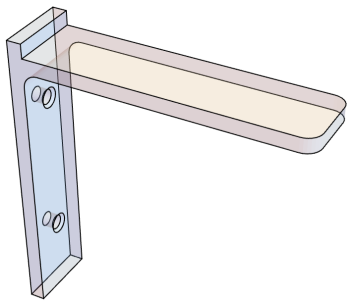
Input cells: 0-30, 114-123.

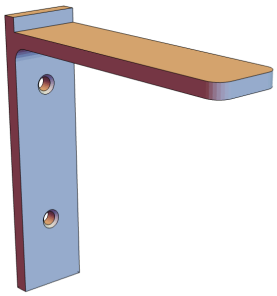
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 03

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(62/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

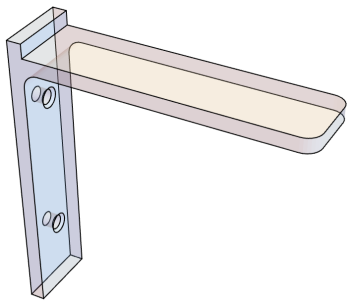
Input cells: 0-30, 114-123.

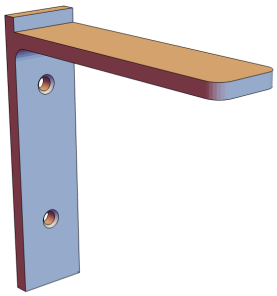
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 04

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(63/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

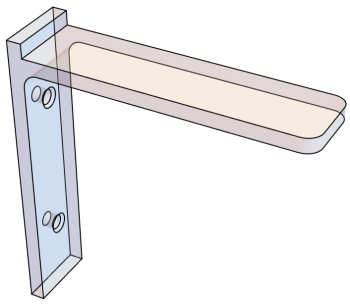
Input cells: 0-30, 114-123.

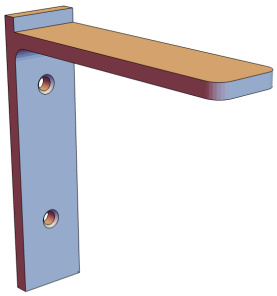
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 05

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(64/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

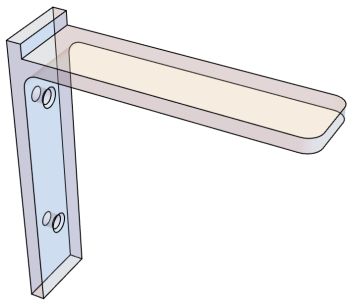
Input cells: 0-30, 114-123.

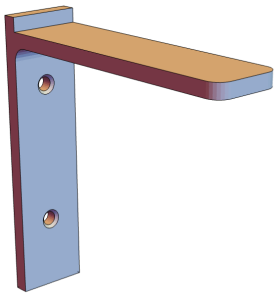
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 06

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(65/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

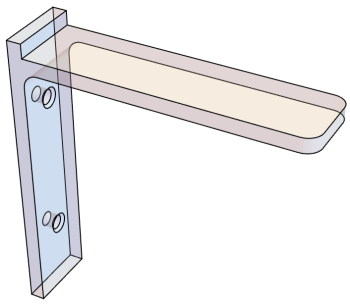
Input cells: 0-30, 114-123.

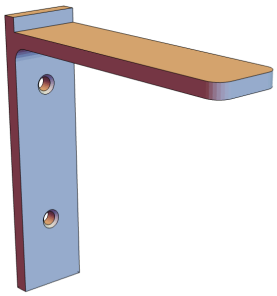
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 07

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(66/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

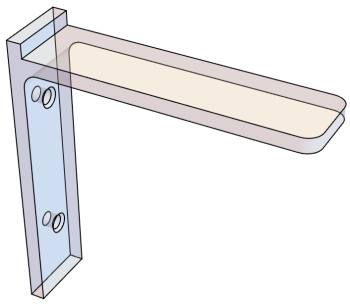
Input cells: 0-30, 114-123.

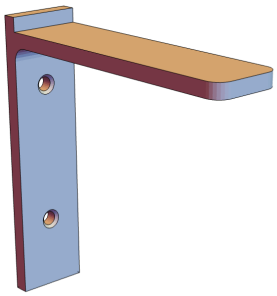
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 08

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(67/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

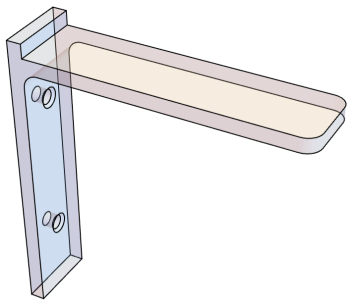
Input cells: 0-30, 114-123.

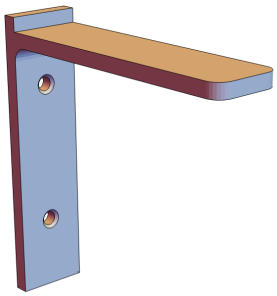
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 09

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(68/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

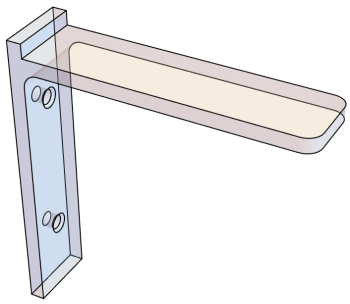
Input cells: 0-30, 114-123.

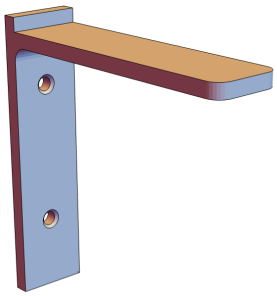
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 10

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(69/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

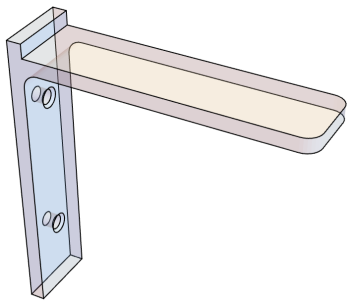
Input cells: 0-30, 114-123.

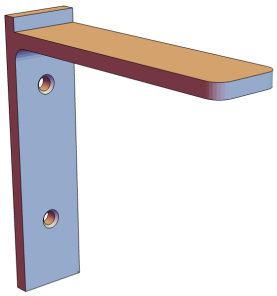
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 11

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(70/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

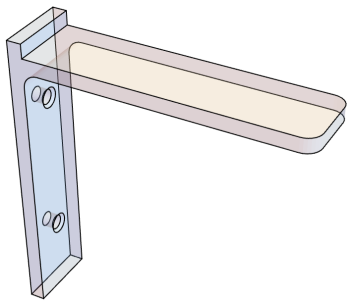
Input cells: 0-30, 114-123.

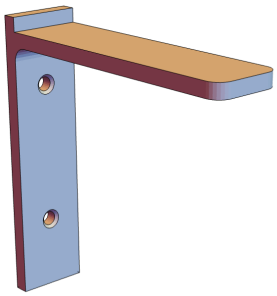
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 12

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(71/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

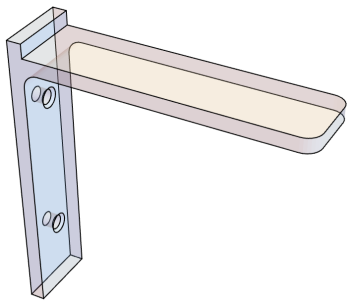
Input cells: 0-30, 114-123.

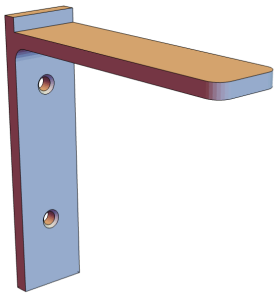
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 13

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(72/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

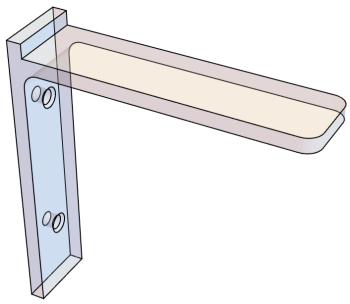
Input cells: 0-30, 114-123.

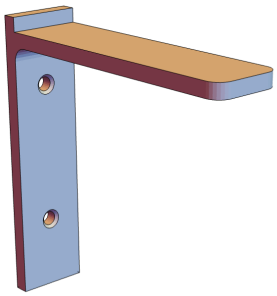
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 14

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(73/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

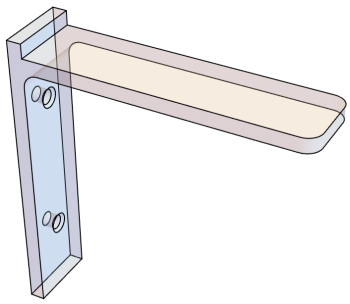
Input cells: 0-30, 114-123.

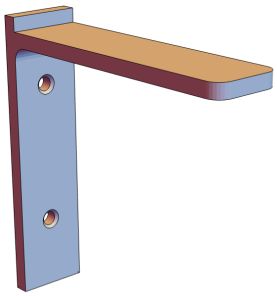
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 15

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(74/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

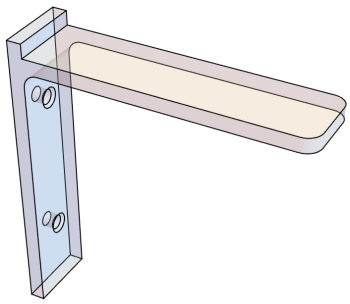
Input cells: 0-30, 114-123.

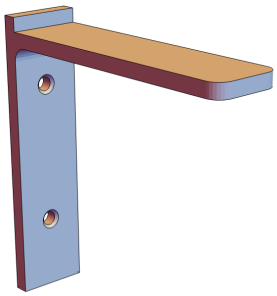
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 16

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(75/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

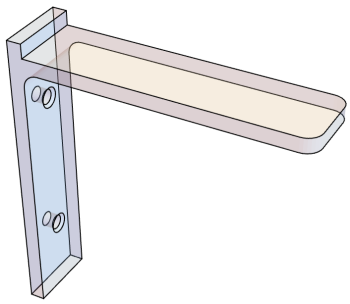
Input cells: 0-30, 114-123.

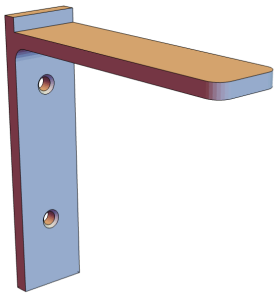
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 17

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(76/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

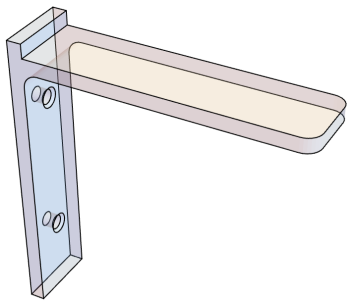
Input cells: 0-30, 114-123.

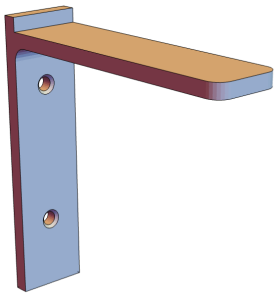
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 18

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(77/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

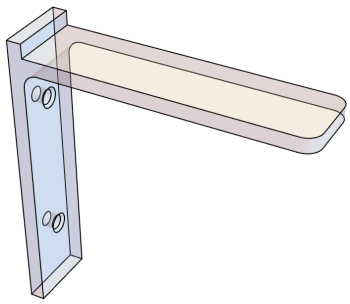
Input cells: 0-30, 114-123.

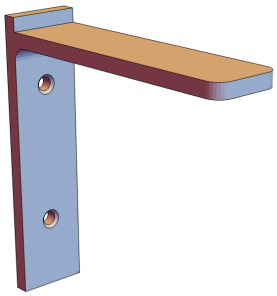
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 19

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(78/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

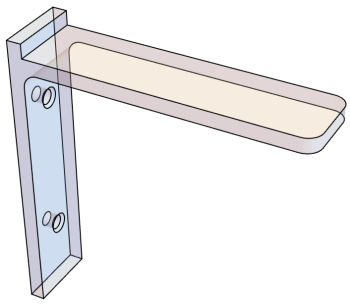
Input cells: 0-30, 114-123.

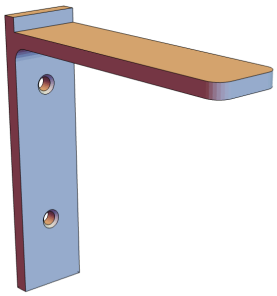
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 20

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(79/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

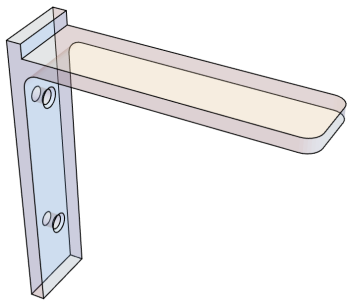
Input cells: 0-30, 114-123.

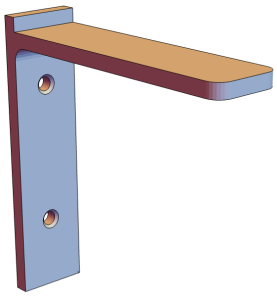
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 21

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(80/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

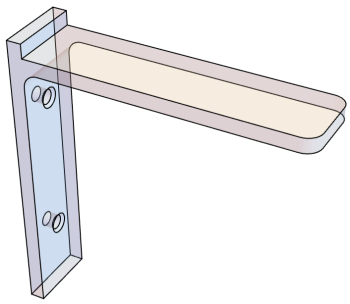
Input cells: 0-30, 114-123.

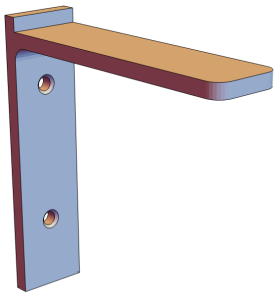
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 22

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(81/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

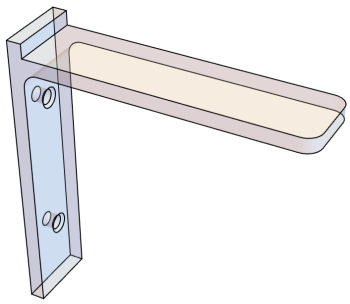
Input cells: 0-30, 114-123.

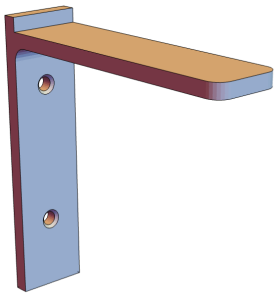
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 23

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(82/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

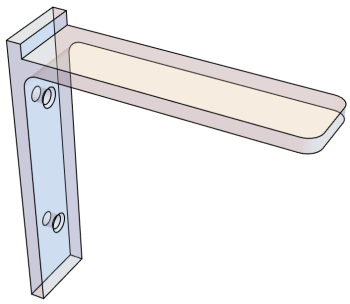
Input cells: 0-30, 114-123.

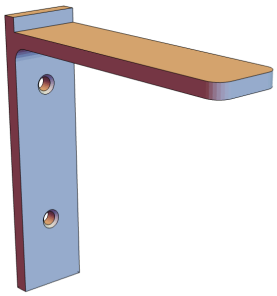
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 24

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(83/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

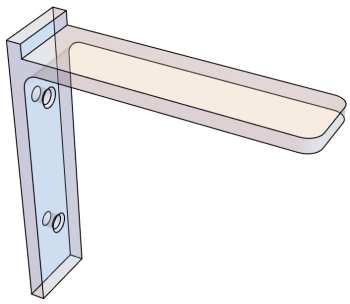
Input cells: 0-30, 114-123.

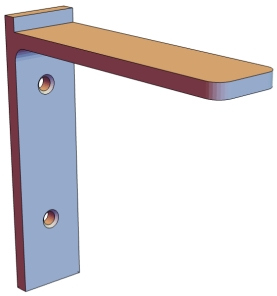
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 25

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(84/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

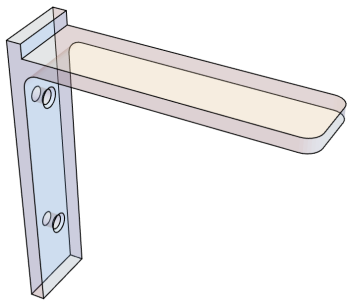
Input cells: 0-30, 114-123.

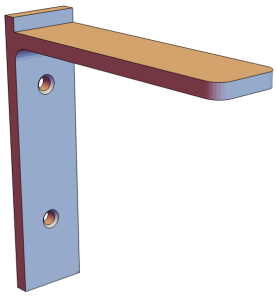
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 26

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(85/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

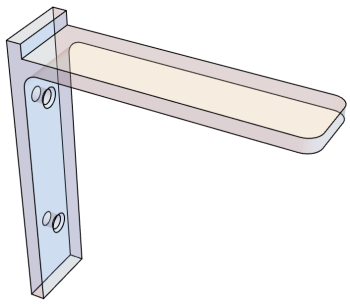
Input cells: 0-30, 114-123.

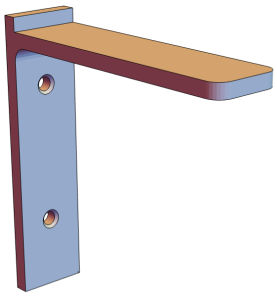
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 27

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(86/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

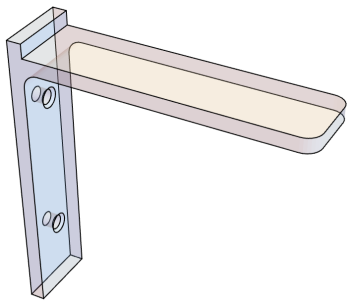
Input cells: 0-30, 114-123.

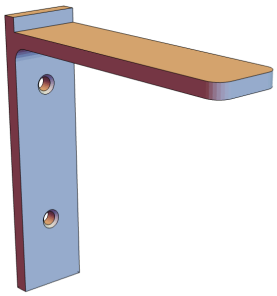
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 28

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(87/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

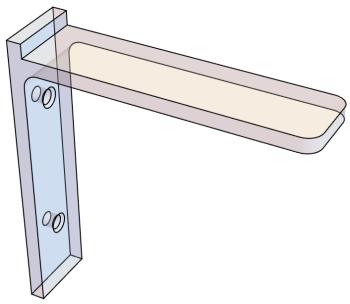
Input cells: 0-30, 114-123.

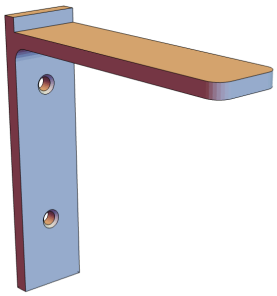
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 29

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(88/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

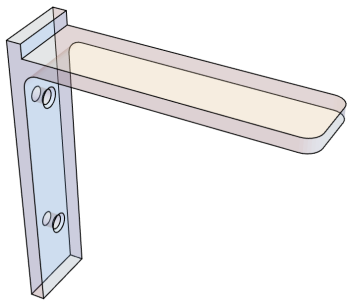
Input cells: 0-30, 114-123.

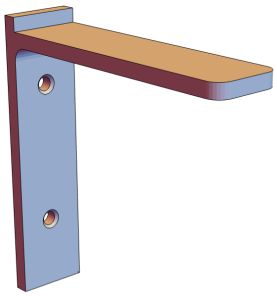
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 30

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(89/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

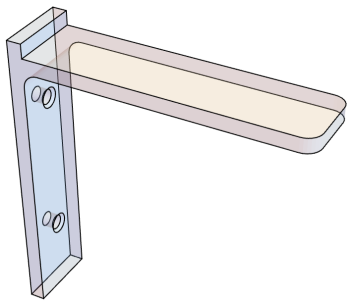
Input cells: 0-30, 114-123.

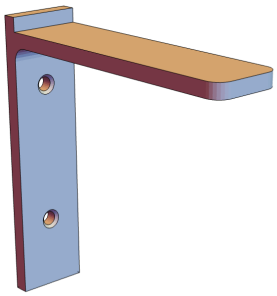
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source harmonic case 31

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the source bracket. Solve this published frequency-sweep member with the exact source frequency-dependent load expression, including its sine factor; do not substitute a constant-amplitude force.

CASE SELECTION

angular_frequency = 10^(90/20) rad/s. Source sweep: Table[10.^i/(2 Pi), {i, 3, 4.5, 0.05}].

REQUIRED RESULTS

Complex displacement/stress fields and the source query-point displacement measure; state complex-amplitude conventions.

SOURCE BINDING

[13] Solid Mechanics

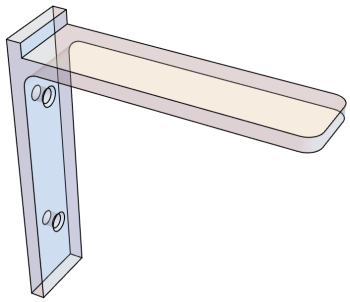
Input cells: 0-30, 114-123.

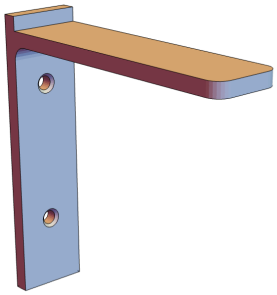
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: transient sinusoidal loading

HARDEST (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the bracket and reproduce the complete source time-dependent load case, initial conditions and mounting conditions.

CASE SELECTION

Time-Dependent Analysis

REQUIRED RESULTS

Query-point displacement and stress histories through the source end time.

SOURCE BINDING

[13] Solid Mechanics

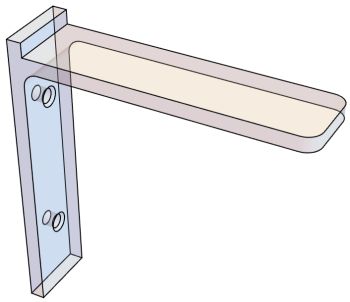
Input cells: 0-30, 82-94.

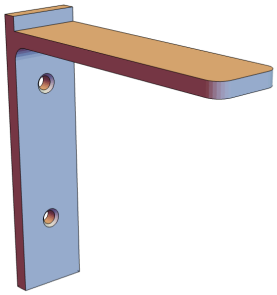
Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Bookshelf bracket: source Poisson-ratio comparison

COMPLEX (proposed) | Family: bookshelf-bracket | Split: bookshelf-bracket





Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the bracket and reproduce the source normalized-material comparison as written. Keep its normalization explicit; do not interpret its unit Young modulus as titanium.

CASE SELECTION

PoissonRatio: 0.1, 0.3, 0.4 (1). Run the complete listed comparison.

REQUIRED RESULTS

Displacement and stress comparison for all three documented Poisson ratios.

SOURCE BINDING

[13] Solid Mechanics

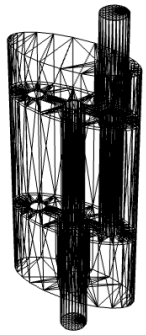
Input cells: 0-30, 103-113.

Geometry construction: source [14]

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Classical muffler: 100 Hz

COMPLEX (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the classical muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Classical: frequency = 100 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

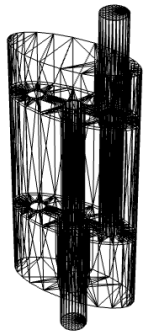
[21] Automotive Muffler Simulation

Input cells: 0-11, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Classical muffler: 200 Hz

COMPLEX (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the classical muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Classical: frequency = 200 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

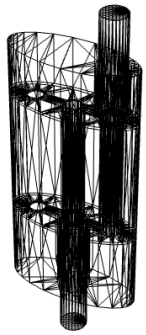
[21] Automotive Muffler Simulation

Input cells: 0-11, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Classical muffler: 300 Hz

COMPLEX (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the classical muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Classical: frequency = 300 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

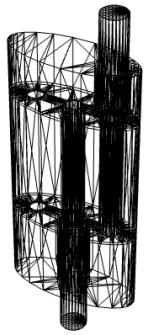
[21] Automotive Muffler Simulation

Input cells: 0-11, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Classical muffler: 400 Hz

COMPLEX (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the classical muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Classical: frequency = 400 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

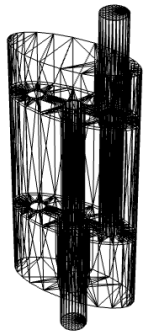
[21] Automotive Muffler Simulation

Input cells: 0-11, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Classical muffler: 500 Hz

COMPLEX (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the classical muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Classical: frequency = 500 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

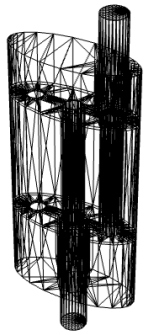
[21] Automotive Muffler Simulation

Input cells: 0-11, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Classical muffler: 600 Hz

COMPLEX (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the classical muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Classical: frequency = 600 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

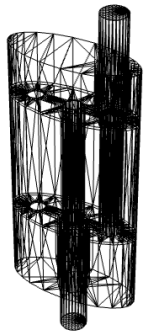
[21] Automotive Muffler Simulation

Input cells: 0-11, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Classical muffler: 700 Hz

COMPLEX (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the classical muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Classical: frequency = 700 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

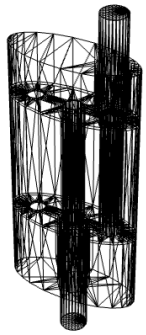
[21] Automotive Muffler Simulation

Input cells: 0-11, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Classical muffler: 800 Hz

COMPLEX (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the classical muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Classical: frequency = 800 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

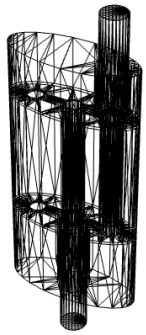
[21] Automotive Muffler Simulation

Input cells: 0-11, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Classical muffler: 900 Hz

COMPLEX (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the classical muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Classical: frequency = 900 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

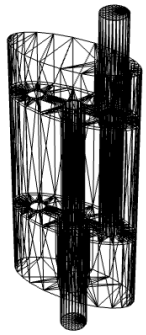
[21] Automotive Muffler Simulation

Input cells: 0-11, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Classical muffler: 1000 Hz

COMPLEX (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the classical muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Classical: frequency = 1000 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

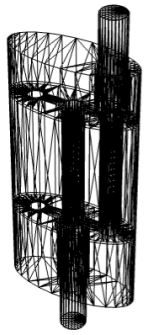
[21] Automotive Muffler Simulation

Input cells: 0-11, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Perforated muffler: 100 Hz

HARDEST (proposed) | Family: muffler-perforated | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the perforated muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Perforated: frequency = 100 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

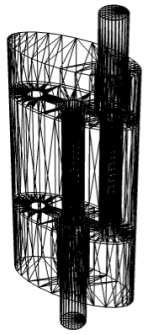
[21] Automotive Muffler Simulation

Input cells: 0-16, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Perforated muffler: 200 Hz

HARDEST (proposed) | Family: muffler-perforated | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the perforated muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Perforated: frequency = 200 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

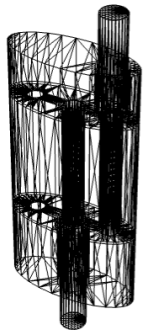
[21] Automotive Muffler Simulation

Input cells: 0-16, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Perforated muffler: 300 Hz

HARDEST (proposed) | Family: muffler-perforated | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the perforated muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Perforated: frequency = 300 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

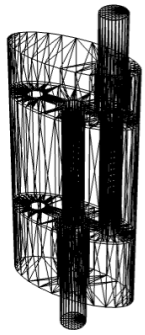
[21] Automotive Muffler Simulation

Input cells: 0-16, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Perforated muffler: 400 Hz

HARDEST (proposed) | Family: muffler-perforated | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the perforated muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Perforated: frequency = 400 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

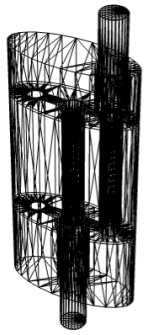
[21] Automotive Muffler Simulation

Input cells: 0-16, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Perforated muffler: 500 Hz

HARDEST (proposed) | Family: muffler-perforated | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the perforated muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Perforated: frequency = 500 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

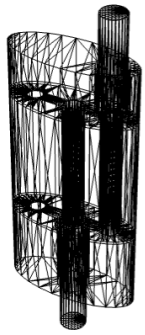
[21] Automotive Muffler Simulation

Input cells: 0-16, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Perforated muffler: 600 Hz

HARDEST (proposed) | Family: muffler-perforated | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the perforated muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Perforated: frequency = 600 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

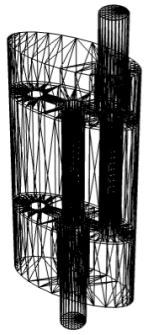
[21] Automotive Muffler Simulation

Input cells: 0-16, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Perforated muffler: 700 Hz

HARDEST (proposed) | Family: muffler-perforated | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the perforated muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Perforated: frequency = 700 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

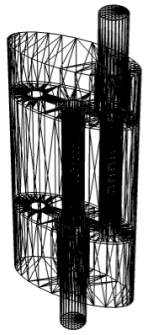
[21] Automotive Muffler Simulation

Input cells: 0-16, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Perforated muffler: 800 Hz

HARDEST (proposed) | Family: muffler-perforated | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the perforated muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Perforated: frequency = 800 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

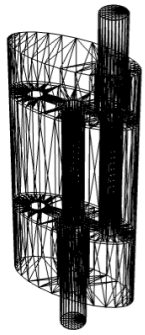
[21] Automotive Muffler Simulation

Input cells: 0-16, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Perforated muffler: 900 Hz

HARDEST (proposed) | Family: muffler-perforated | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the perforated muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Perforated: frequency = 900 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

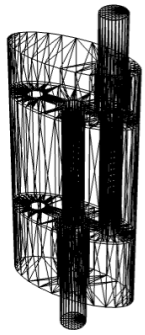
[21] Automotive Muffler Simulation

Input cells: 0-16, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Perforated muffler: 1000 Hz

HARDEST (proposed) | Family: muffler-perforated | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the perforated muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Perforated: frequency = 1000 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

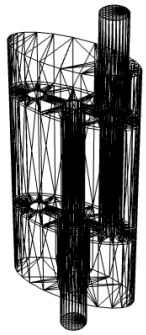
[21] Automotive Muffler Simulation

Input cells: 0-16, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Padded muffler: 100 Hz

HARDEST (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the padded muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Padded: frequency = 100 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

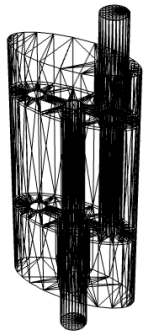
[21] Automotive Muffler Simulation

Input cells: 0-11, 17-22, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Padded muffler: 200 Hz

HARDEST (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the padded muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Padded: frequency = 200 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

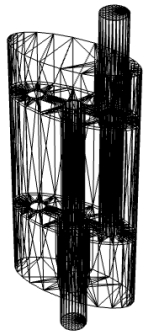
[21] Automotive Muffler Simulation

Input cells: 0-11, 17-22, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Padded muffler: 300 Hz

HARDEST (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the padded muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Padded: frequency = 300 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

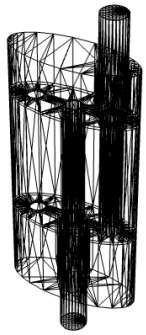
[21] Automotive Muffler Simulation

Input cells: 0-11, 17-22, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Padded muffler: 400 Hz

HARDEST (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the padded muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Padded: frequency = 400 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

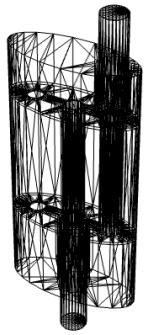
[21] Automotive Muffler Simulation

Input cells: 0-11, 17-22, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Padded muffler: 500 Hz

HARDEST (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the padded muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Padded: frequency = 500 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

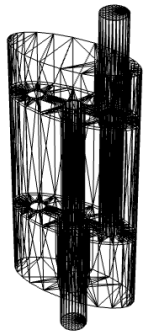
[21] Automotive Muffler Simulation

Input cells: 0-11, 17-22, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Padded muffler: 600 Hz

HARDEST (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the padded muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Padded: frequency = 600 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

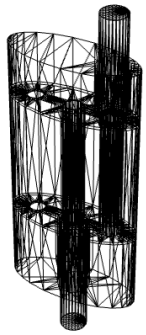
[21] Automotive Muffler Simulation

Input cells: 0-11, 17-22, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Padded muffler: 700 Hz

HARDEST (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the padded muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Padded: frequency = 700 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

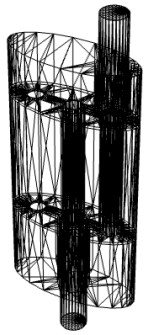
[21] Automotive Muffler Simulation

Input cells: 0-11, 17-22, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Padded muffler: 800 Hz

HARDEST (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the padded muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Padded: frequency = 800 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

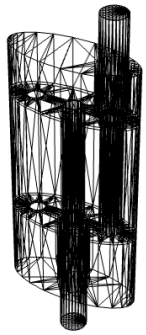
[21] Automotive Muffler Simulation

Input cells: 0-11, 17-22, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Padded muffler: 900 Hz

HARDEST (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the padded muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Padded: frequency = 900 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

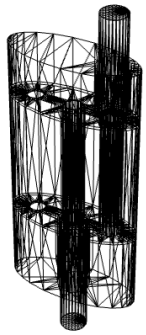
[21] Automotive Muffler Simulation

Input cells: 0-11, 17-22, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Padded muffler: 1000 Hz

HARDEST (proposed) | Family: muffler-classical | Split: automotive-mufflers



Source engineering geometry. Click an image to open its original.

CAD + FEM TASK

Reconstruct the padded muffler air volume from the source support geometry, retaining internal partitions and apertures. Solve the selected source frequency. For the padded case also reconstruct its material subregions and use its empirical complex coefficients.

CASE SELECTION

Padded: frequency = 1000 Hz. Source sweep: Table[f, {f, 20, 1000, 10}].

REQUIRED RESULTS

Complex acoustic pressure and source-defined transmission loss, using the documented outlet probe and power convention.

SOURCE BINDING

[21] Automotive Muffler Simulation

Input cells: 0-11, 17-22, 27-29.

Apply all shared requirements on page 2. Dependency resolution, solver validation and model-failure testing remain pending.

Sources 1-7

[01] Thermal and Structural Analysis of a Disc Brake

https://reference.wolfram.com/language/PDEModels/tutorial/HeatTransfer/ModelCollection/DiscBrake.html

Full code cells and cached-page SHA256: sources.json.

[02] Shrink Fitting of Assembly

https://reference.wolfram.com/language/PDEModels/tutorial/HeatTransfer/ModelCollection/ShrinkFitting.html

Full code cells and cached-page SHA256: sources.json.

[03] Contactless Anemometer

https://reference.wolfram.com/language/PDEModels/tutorial/HeatTransfer/ModelCollection/Anemometer.html

Full code cells and cached-page SHA256: sources.json.

[04] Laminar Flow

https://reference.wolfram.com/language/PDEModels/tutorial/FluidDynamics/LaminarFlow.html

Full code cells and cached-page SHA256: sources.json.

[05] 3D Print and Mechanical Design

https://reference.wolfram.com/language/PDEModels/tutorial/StructuralMechanics/ModelCollection/3DPrintedMechanicalDesign.html

Full code cells and cached-page SHA256: sources.json.

[06] Hyperelastic Model Comparison

https://reference.wolfram.com/language/PDEModels/tutorial/StructuralMechanics/ModelCollection/HyperelasticModelComparison.html

Full code cells and cached-page SHA256: sources.json.

[07] Biaxial Tensile Test of Hyperelastic Tissue

https://reference.wolfram.com/language/PDEModels/tutorial/StructuralMechanics/ModelCollection/BiaxialTensileTestHyperelasticTissue.html

Full code cells and cached-page SHA256: sources.json.

Scope note: mfem.org was crawled (150 pages), but no case from that crawl met the selected combined CAD/FEM/engineering-drawing criteria. This catalog does not add physical setups to geometry-only examples.

Sources 8-14

[08] Helmholtz Resonator

https://reference.wolfram.com/language/PDEModels/tutorial/Acoustics/ModelCollection/HelmholtzResonator.html

Full code cells and cached-page SHA256: sources.json.

[09] Quasistatic Magnetic Fields

https://reference.wolfram.com/language/PDEModels/tutorial/Electromagnetics/QuasistaticMagneticFields.html

Full code cells and cached-page SHA256: sources.json.

[10] Electrostatically Actuated MEMS Device

https://reference.wolfram.com/language/PDEModels/tutorial/Multiphysics/ModelCollection/ElectrostaticallyActuatedMEMS.html

Full code cells and cached-page SHA256: sources.json.

[11] Hygroscopic Swelling

https://reference.wolfram.com/language/PDEModels/tutorial/Multiphysics/ModelCollection/HygroscopicSwelling.html

Full code cells and cached-page SHA256: sources.json.

[12] Water Condensation of Passive Dew Condensers

https://reference.wolfram.com/language/PDEModels/tutorial/Multiphysics/ModelCollection/PassiveDewCondenser.html

Full code cells and cached-page SHA256: sources.json.

[13] Solid Mechanics

https://reference.wolfram.com/language/PDEModels/tutorial/StructuralMechanics/SolidMechanics.html

Full code cells and cached-page SHA256: sources.json.

[14] Book Shelf Bracket

https://reference.wolfram.com/language/OpenCascadeLink/tutorial/BookShelfBracket.html

Full code cells and cached-page SHA256: sources.json.

Scope note: mfem.org was crawled (150 pages), but no case from that crawl met the selected combined CAD/FEM/engineering-drawing criteria. This catalog does not add physical setups to geometry-only examples.

Sources 15-21

[15] Solid Mechanics Model Verification Tests

https://reference.wolfram.com/language/PDEModels/tutorial/StructuralMechanics/SolidMechanicsVerificationTests.html

Full code cells and cached-page SHA256: sources.json.

[16] Plasticity

https://reference.wolfram.com/language/PDEModels/tutorial/StructuralMechanics/Plasticity.html

Full code cells and cached-page SHA256: sources.json.

[17] Magnetostatics for Permanent Magnets

https://reference.wolfram.com/language/PDEModels/tutorial/Electromagnetics/MagnetostaticsForPermanentMagnets.html

Full code cells and cached-page SHA256: sources.json.

[18] Permanent Magnet with Coin

https://reference.wolfram.com/language/OpenCascadeLink/tutorial/PermanentMagnetWithCoin.html

Full code cells and cached-page SHA256: sources.json.

[19] Heat Transfer Model Verification Tests

https://reference.wolfram.com/language/PDEModels/tutorial/HeatTransfer/HeatTransferVerificationTests.html

Full code cells and cached-page SHA256: sources.json.

[20] Joule Heating of Tungsten Wire

https://reference.wolfram.com/language/PDEModels/tutorial/Multiphysics/ModelCollection/JouleHeating.html

Full code cells and cached-page SHA256: sources.json.

[21] Automotive Muffler Simulation

https://reference.wolfram.com/language/PDEModels/tutorial/Acoustics/ModelCollection/AcousticMuffler.html

Full code cells and cached-page SHA256: sources.json.

Scope note: mfem.org was crawled (150 pages), but no case from that crawl met the selected combined CAD/FEM/engineering-drawing criteria. This catalog does not add physical setups to geometry-only examples.